GEOG7305 Programming for Geospatial Data

Lab02 Entity Relationship Diagram

Lab Deliverables:

Draw ER diagram which is based on your Lab01 tree management system. Please append your ERD at the end your Lab01 report. Thus, your lab02 report should include: description of the DFD, DFD, ERD, and brief description of your ERD.

Exercise1: Draw ER-Diagram

The University of Hong Kong has several departments. Each department is managed by a chair, and at least a professor. *Professors must be assigned to one, but possibly more departments*. At least one professor teaches a course. Each course may be taught more than once by different professors. We know of the department name, the professor name, the professor employee id, the course names, the course schedule, the term/year that the course is taught, the departments the professor is assigned to, the department that offers the course.

Please draw the ER-diagram and define the database table.

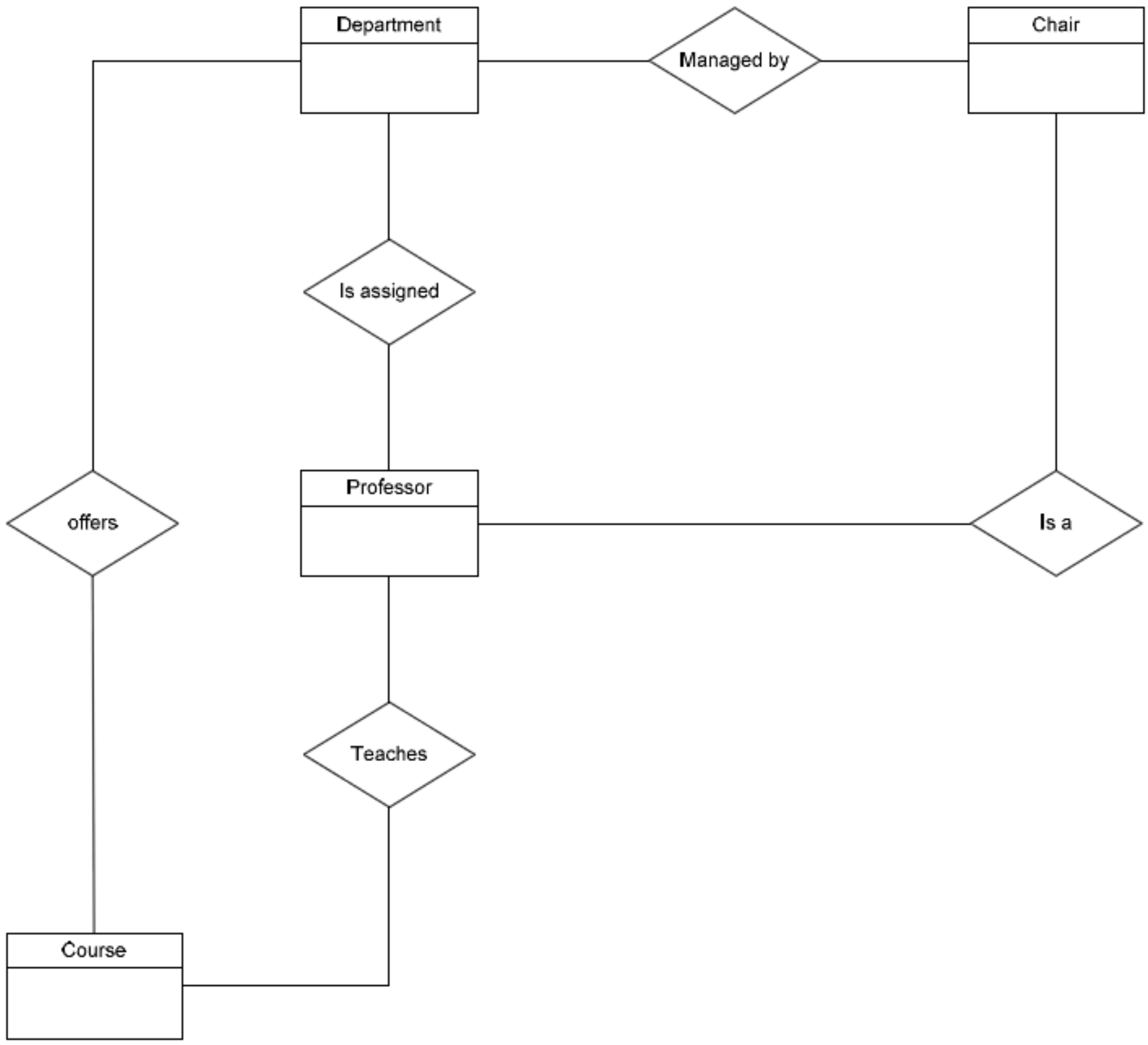
Ex1.1 List out the Entities:

Identify Relationships (fill in the blank with *):

	Department	Chair	Professor	Course
department		*	*	*
chair	*		*	
Professor	*			*
course	*		*	

Ex1.2 Draw the entities and relationship (rough)

Please fill in the Cardinality of the following diagram

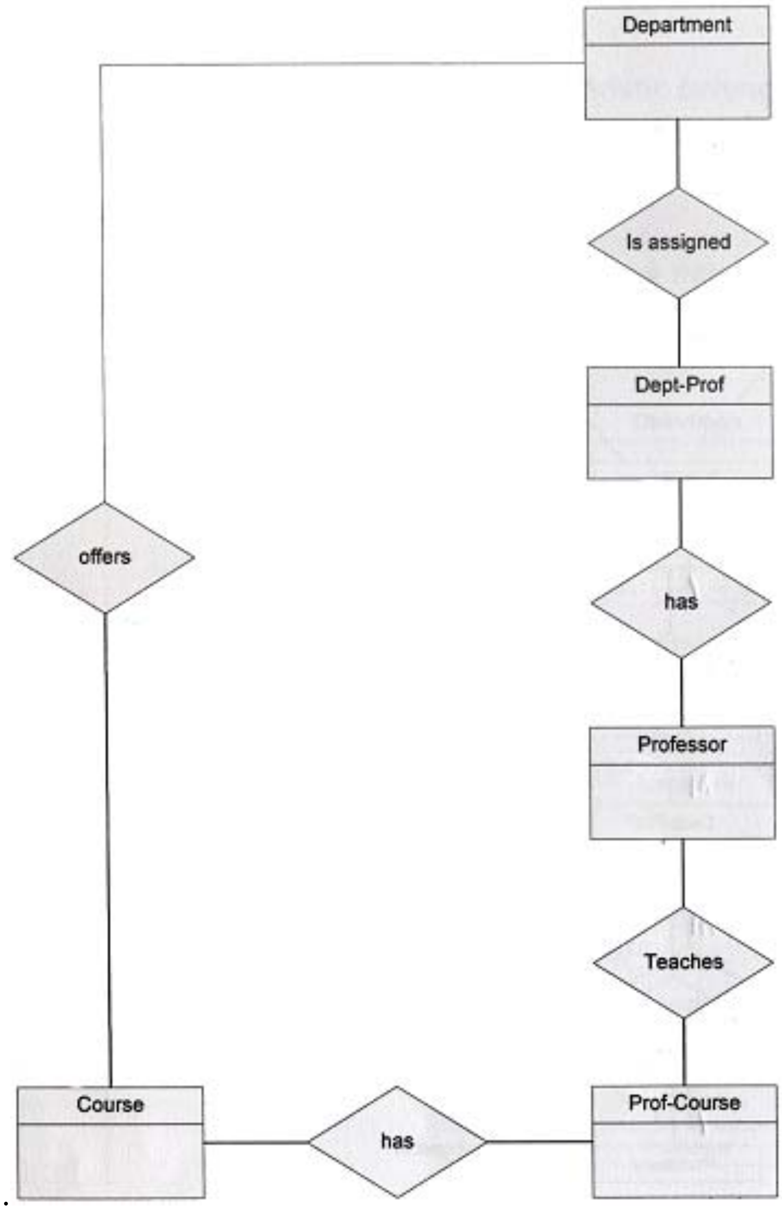


Ex1.3 Eliminate many-many relationships, and collapse one-to-one relationships

Since Professors must be assigned to one, but possibly more departments. Relationship between Department and Professor is M to N. Eliminate it by make creating another composite/bridge entity.\name as **Dept-Prof**. Then link them by two 1-M relationship.

Also eliminate the M-N relationship between Professor and Course. Eliminate it by introduce add **Prof-Course** bridge entity

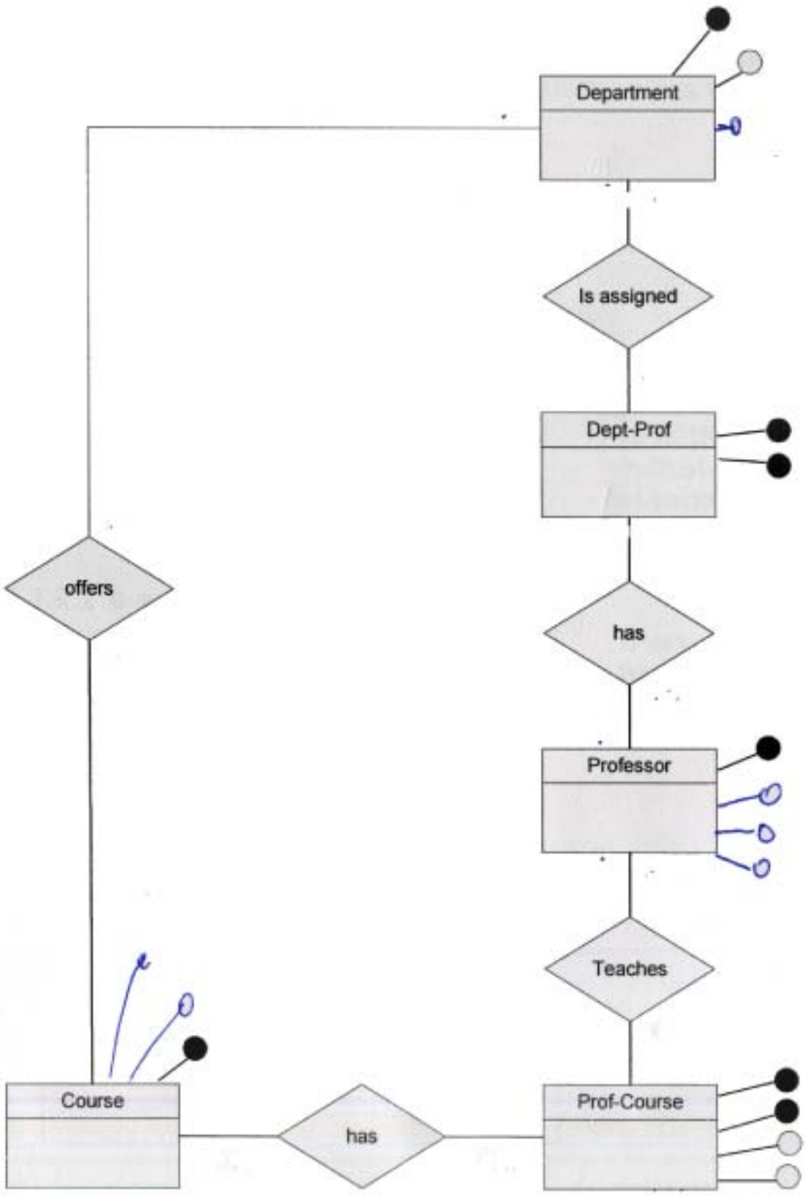
The following diagram has eliminate the M-N relationships, *please file in the cardinality*:



Ex1.4 Define Primary Keys (fill in the blank)

Entity	Primary Key
Department	
Dept-prof	
Professor	
Prof-course	
Course	

Ex1.5 Full ER diagram(fill in the cardinality and attribute)



Ex1.6 Finally Define Database Tables:

Department

Dept ID	Dept Name	Chair employee ID

Dept-Prof

Dept ID	**Employee ID**	Assign date

Professor

Employee ID	Name	Tel	address

Prof-Course

Employee ID	**Course ID**	Schedule	Term

Course

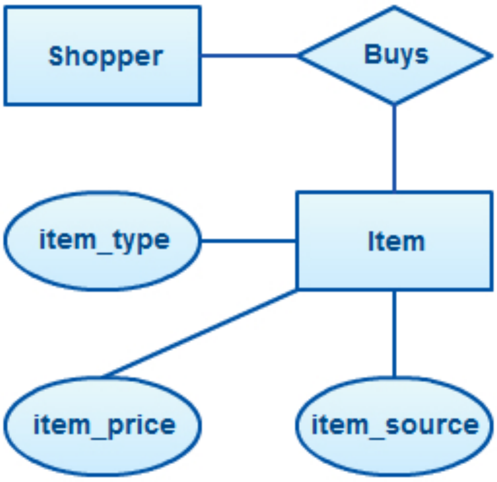
Course ID	Course Name	Description

Appendix:

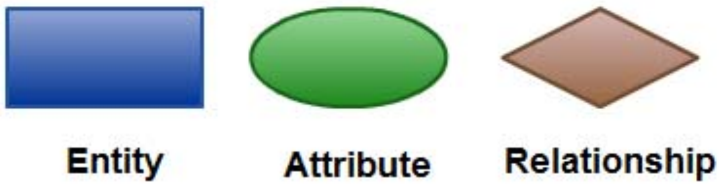
Entity Relationship Diagram

An Entitity Relationship Diagram (ERD) is a visual representation of different data using conventions that describe how these data are related to each other. ERD technique is a graphic way of displaying entity types. Relationship types, and attributes. In particular, ER diagrams are frequently used during the design stage of a development process in order to identify different system elements and their relationships with each other.

Simple ERD



ER Diagram Symbols and Notations

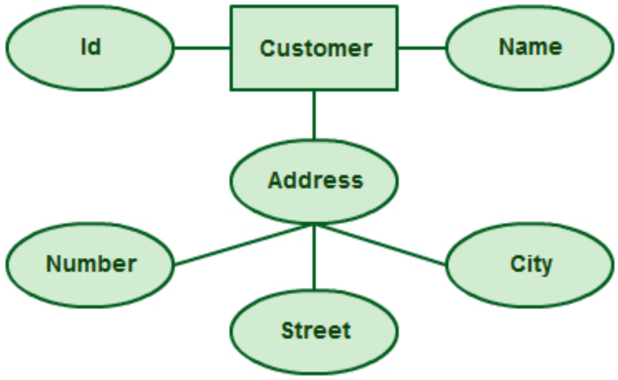


Entity

An entity can be a person, place, event, or object that is relevant to a given system. For example, a school system may include students, teachers, major courses, subjects, fees, and other items. Entities are represented in ER diagrams by a rectangle and named using singular nouns.

Attribute

An attribute is a property, trait, or characteristic of an entity, relationship, or another attribute. For example, the attribute Inventory Item Name is an attribute of the entity Inventory Item. An entity can have as many attributes as necessary. Meanwhile, attributes can also have their own specific attributes. For example, the attribute “customer address” can have the attributes number, street, city, and state. These are called composite attributes. Note that some top level ER diagrams do not show attributes for the sake of simplicity. In those that do, however, attributes are represented by oval shapes.



Relationship

A relationship describes how entities interact. For example, the entity "employee" may be related to the entity "department" by the relationship "work for". Relationships are represented by diamond shapes and are labeled using verbs.

Cardinality and Ordinality

These two further defines relationships between entities by placing the relationship in the context of numbers. In an email system, for example, one account can have multiple contacts. The relationship in this case follows a "one to many" model. There are number of notations used to present cardinality in ER diagrams.

One to one (1:1) One to many (1:m) Many to one (m:1) Many to many (m:n)

- one-to-one

- one to many

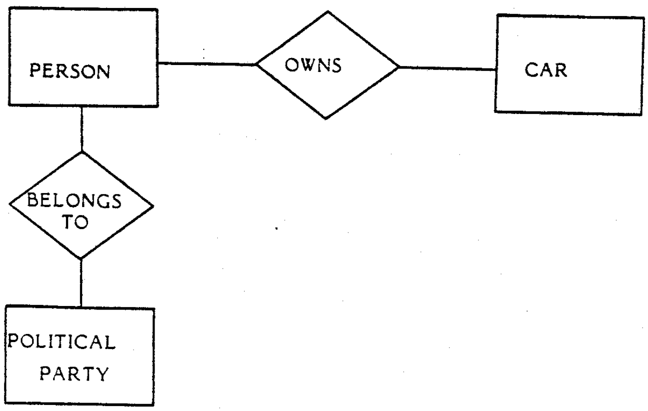
- many-to-many

Tips on How to Draw ER Diagrams

1. **Identify all the relevant entities** in a given system and determine the **relationships** among these entities.
2. An entity should **appear only once** in a particular diagram.
3. Provide a precise and **appropriate name** for each entity, attribute, and relationship in the diagram. Terms that are simple and familiar always beats vague, technical-sounding words. In naming entities, remember to use singular nouns. However, adjectives may be used to distinguish entities belonging to the same class (part-time employee and full time employee, for example). Meanwhile attribute names must be meaningful, unique, system- independent, and easily understandable. (**entities◊nouns, relationship◊verb**)
4. Remove vague, redundant or unnecessary relationships between entities.
5. Never connect a relationship to another relationship.
6. Make effective use of colors. You can use colors to classify similar entities or to highlight key areas in your diagrams.

Example:

English sentence: A person may own a car may belong to a political party
Nouns(entity): person, car, political party
Verbs(relationship): own, belong to



Solving M-N relationship Can be implemented by breaking it up to produce a set of 1:M relationships. By creating a composite entity or ***bridge entity***. This will be used to link the

tables that were originally related in a M:N relationship. The composite entity structure includes- as foreign keys-at least the primary keys of the tables that are to be linked.

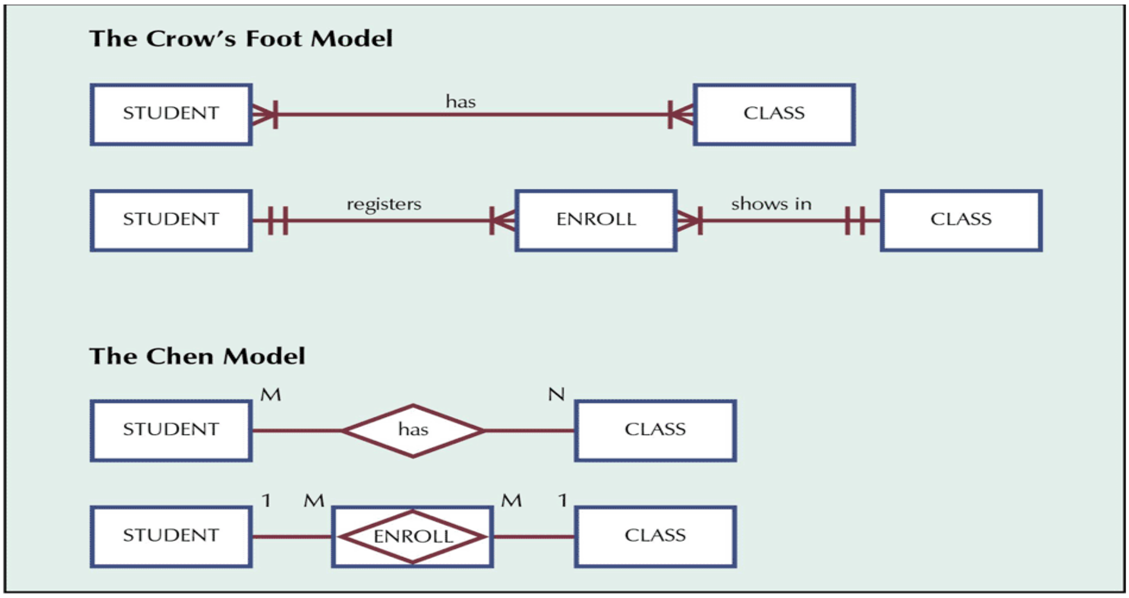


Table name: STUDENT
Primary key: STU_NUM
Foreign key: none

Database name: Ch03_CollegeTry2

STU_NUM	STU_LNAME
321452	Bowser
324257	Smithson

Table name: ENROLL
Primary key: CLASS_CODE + STU_NUM
Foreign key: CLASS_CODE, STU_NUM

CLASS_CODE	STU_NUM	ENROLL_GRADE
10014	321452	C
10014	324257	B
10018	321452	A
10018	324257	B
10021	321452	C
10021	324257	C

Table name: CLASS
Primary key: CLASS_CODE
Foreign key: CRS_CODE

CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	CLASS_ROOM	PROF_NUM
10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
10018	CIS-220	2	MWF 9:00-9:50 a.m.	KLR211	114
10021	QM-261	1	MWF 8:00-8:50 a.m.	KLR200	114

Entity: Student, Class → M-N relationship
Bridge entity: Enroll → that help to break M-N to two 1-M relationship